Ahmed Elbialy

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Summary

Backend-focused Full-Stack Engineer with 2+ years of experience building scalable web applications, APIs, system integrations, and data pipelines using Next.js, FastAPI, PostgreSQL, AWS, and GCP.
Specialized in AI agent engineering, building LangGraph/LangChain-based systems with tool integrations, execution control, and multi-step orchestration for intelligent automation.
Experienced in business-system integrations, Google Apps Script and clasp development, secure cloud services, and performance optimization across distributed applications.

Experience

Modern Intelligence Solutions (MIS) **July 2025 – Present**
Full-Stack Software Engineer (Backend-Focused) *On-Site, Al Shorouk City, Cairo, Egypt*

- Architected and developed scalable full-stack applications using Next.js, with a strong focus on backend system design, API architecture, and performance optimization.
- Leveraged AI-driven development workflows using Cursor IDE to significantly accelerate development speed, debugging efficiency, and code quality.
- Designed and implemented AI agents using LangGraph and LangChain SDK, building tool-integrated systems (e.g., Tavily web search) with controlled execution flows, interrupts, and node-based orchestration loops.
- Engineered intelligent multi-step agent pipelines capable of dynamic decision-making and task automation across complex workflows.
- Engineered a multi-layer agentic RAG system using PostgreSQL/pgvector, full-text search, and reciprocal rank fusion, enabling source-cited answers grounded in project documents and reference materials.
- Built Python document-ingestion workflows for PDF, DOCX, screenplay, and budget formats, integrating Google Document AI OCR fallback, Voyage AI embeddings, semantic chunking, and resumable PostgreSQL jobs.
- Implemented LangGraph conversation-context handling with token-aware history summarization, prompt caching, and retry logic to support lengthy user interactions and unreliable LLM API responses.
- Architected and deployed production-grade cloud infrastructure for AI agent systems on AWS, orchestrating containerized services via ECS and ECR, implementing load balancing, secure networking (SSL/TLS), and managed databases (RDS PostgreSQL) to ensure scalability and high availability.
- Developed application services integrated with a **GCP staging environment** using Cloud Run v2, Cloud SQL (PostgreSQL/pgvector), Memorystore Redis, GCS, Secret Manager, and Cloud Armor-protected HTTPS access.
- Designed and managed PostgreSQL databases on Supabase, implementing robust data models, secure RBAC systems, and high-performance API endpoints.
- Implemented Firebase authentication for a Next.js application, including TOTP/SMS multi-factor authentication, PostgreSQL-backed sessions, token-revocation checks, and account lockout controls.
- Engineered integrations with third-party platforms (Paychex, AxisCare, Relias), building reliable authentication layers and data synchronization pipelines across distributed systems.
- Developed automation systems using Python (Selenium, Playwright, Pandas, XLWings) to extract, process, and transform large datasets, reducing manual operational overhead.
- Built asynchronous, event-driven backend services using FastAPI and Celery, enabling scalable task processing, background jobs, and data pipelines.
- Built a Next.js server-sent events (SSE) chat integration that streams LangGraph agent responses to the browser, persists messages, and surfaces inline citations for source verification.
- Designed and developed a cross-platform desktop application using PyQt, implementing structured UI architecture, local data persistence, and workflow automation.
- Developed Google Apps Script solutions to populate Google Sheets and send emails programmatically, using **clasp** for local development and version control with Git.

Anyware Software **Aug 2024 – June 2025**
Cloud Full-Stack Software Engineer *On-Site, Maadi, Cairo, Egypt*

- Built and deployed cloud-native, serverless applications using AWS Amplify, Lambda, Cognito, S3, DynamoDB, and GraphQL APIs, ensuring high availability and scalability.
- Designed and developed responsive web and mobile applications using React and React Native (Expo), delivering consistent cross-platform user experiences.
- Collaborated in production environments to deliver scalable frontend-backend integrations with optimized performance and maintainability.

Digital Car **Aug 2024 – Oct 2025**
Backend Node.js Developer (Part-Time) *Remote, Cairo, Egypt*

- Developed and maintained scalable backend services using NestJS and MongoDB, optimizing database schemas and improving system performance under production workloads.
- Integrated multiple third-party services including Stripe (payments), Twilio (SMS), SendGrid (email), and AWS S3 (storage), enabling complex financial and transactional workflows.
- Collaborated within a Scrum-based development environment to deliver reliable, production-ready backend systems.

Vultara, Inc. **May 2024 – July 2024**
Software Engineer (Freelance) *Remote, Troy, Michigan, USA*

- Contributed to large-scale MEAN stack applications, handling both frontend (Angular) and backend (Node.js, Express.js) development.
- Improved system performance and scalability while resolving complex full-stack issues in production environments.
- Collaborated with distributed teams to deliver stable, maintainable, and high-quality software solutions.

Freelancing **Sep 2023 – May 2024**
Backend Engineer *Remote*

- Developed full-stack applications including social platforms using GraphQL APIs and React.js, focusing on scalable backend architecture.
- Implemented best practices in version control, CI workflows, and collaborative development using Git and GitHub.

ValleySoft **June 2024 – July 2024**
Integration Engineer Intern *Heliopolis, Cairo*

- Built integration solutions using IBM Cloud Pack tools (App Connect, MQ Explorer), enabling communication between distributed enterprise systems.
- Designed and implemented message transformation pipelines and middleware logic for system interoperability.
- Developed SOAP and REST APIs handling XML and JSON data formats to support enterprise-level integrations.

Technical Skills

Backend & API Development: Node.js (Express, NestJS, Fastify), Python (FastAPI), Spring Boot, REST, GraphQL, SSR
Frontend Development: React.js, Next.js, React Native (Expo), HTML5, CSS3, JavaScript, TypeScript
AI & Agent Engineering: LangChain, LangGraph, LLM Agents, Hybrid RAG, pgvector, Tool Integration, Agent Orchestration, Prompt Engineering
Cloud & DevOps: AWS (ECS, ECR, RDS, S3, Lambda, Cognito, Amplify), GCP (Cloud Run, Cloud SQL, GCS, Memorystore, Secret Manager, Cloud Armor), Docker, Docker Compose, Nginx, Kubernetes, CI/CD
Databases: PostgreSQL, pgvector, Supabase, MongoDB, DynamoDB, Redis
Data & Automation: Google Apps Script, clasp, Google Sheets, Pandas, Selenium, Playwright, XLWings, Document AI, Data Pipelines
Developer Tools: Cursor (AI IDE), VS Code, IntelliJ, Postman, Git, GitHub
Software Engineering: System Design, Distributed Systems, OOP, SOLID Principles

Education

Cairo University **Sep. 2018 – May 2023**
Bachelor of Electrical Power Engineering (GPA: Very Good) *Giza, Giza*

National Telecommunication Institute (NTI) **Feb 2024 – June 2024**
SOA and Microservices Intern *Nasr City, Cairo*

- Completed a hands-on internship focused on SOA, microservices, and BPM using Java, Spring Boot, IBM BAW, and IBM integration tools (App Connect, MQ Explorer, API Connect).
- Gained exposure to DevOps (Docker, Kubernetes, Jenkins, Terraform, Ansible), RPA with IBM tools, and developed strong soft skills in communication, teamwork, and problem-solving.

Languages

Arabic: Native
English: B2 (Upper Intermediate)
German: A1 (Beginner, Currently learning)